Laravel – Database & Migrations

Day 8 Study Notes • MySQL + .env + Migrations + Columns + Commands

1. Database in Laravel

Laravel application data can be stored in a database. In this Day 8 topic, the database used is MySQL.

```
Laravel Application
        ↓
Database Connection
        ↓
MySQL
        ↓
Database
        ↓
Tables
```

2. Database vs Table

Database ■■■■■■ tables-■ ■■■■■■■■■■■■■■■ container. **Table** ■■■■■■ data structure.

```
firstproject  ← Database
    ■
    ■■■ students  ← Table
    ■■■ courses   ← Table
    ■■■ users     ← Table
```

3. .env ■■■■■■■■ ■■■■■?

Laravel project root-■■ ■■■■■■■■■■ **.env** file environment-specific configuration values-■ ■■■■■■■■■■■■■■■■. Database connection settings-■ ■■■■■■ ■■■■■■■■■■■■■.

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=firstproject
DB_USERNAME=root
DB_PASSWORD=
```

DB_CONNECTION → database driver. **DB_HOST** → database server location. **DB_PORT** → MySQL port. **DB_DATABASE** → database name. **DB_USERNAME** → username. **DB_PASSWORD** → password.

Important: .env-■■ DB_DATABASE name ■■■■■■■■■■■ database automatically create ■■■■■■. Database MySQL-■■ ■■■■■■■ ■■■■■■■■■.

4. Migration ■■■■■■■■ ■■■■■?

Migration ■■■■■■■ database structure-■ code ■■■■■■ create/change/manage ■■■■■■ Laravel ■■■■■■■■■ mechanism. Table structure changes-■ project code-■■■■■ track ■■■■■■ migration ■■■■■■■.

```
Migration
    ↓
Database structure instructions
    ↓
php artisan migrate
    ↓
MySQL table structure
```

5. Why Migration?

Database changes-■ manually SQL commands ■■■■■■■ ■■■■■■■■■■■■■■■, repeatable ■■■■■■■■ trackable code ■■ ■■■■■■■■■■■■ migration ■■■■■■■. Migration-■ database structure-■■■■■■ version-controlled change history ■■■■ ■■■■■■■■■■■■.

6. Migration File Location

```
database/
■■■■ migrations/
    ■■■■ 2026_..._create_users_table.php
    ■■■■ 2026_..._create_cache_table.php
    ■■■■ 2026_..._create_students_table.php
```

7. Create a Migration

```
php artisan make:migration create_students_table
```

■■■■■ command migration **file**-■ create ■■■■■■■■■■. ■■■■ database table-■ ■■■■■ create ■■■■■■■■■.

8. Migration File Structure

```
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        // apply change
    }

    public function down(): void
    {
        // undo change
    }
};
```

9. up() ■■■■■■■■ down()

up() → migration run ■■■■■■■■■■■■■■ apply ■■■■■■ ■■■■■■■■■ database change.

down() → migration rollback ■■■■■■■■■■■■■■ ■■■■■■ ■■■■■■■■■ reverse change.

```
up()
 ↓
Create / modify table

rollback
 ↓
down()
 ↓
Undo that change
```

10. Create students Table

```
public function up(): void
{
    Schema::create('students', function (Blueprint $table) {
```

```
        $table->id();
        $table->string('name');
        $table->string('email');
        $table->integer('age');
        $table->timestamps();
    });
}
```

11. Schema::create()

```
Schema::create('students', function (Blueprint $table) {
    // columns
});
```

[redacted] students [redacted] table name. Closure-[redacted] `$table` [redacted] columns define [redacted].

12. Common Columns

```
$table->id();
$table->string('name');
$table->string('email');
$table->integer('age');
$table->timestamps();
```

id() → primary key id column. **string()** → string/varchar-type column. **integer()** → integer column. **timestamps()** → created_at [redacted] updated_at columns.

13. $table->id()

```
$table->id();
```

Students table-[redacted] primary identifier [redacted] id column-[redacted] define [redacted].

14. $table->string()

```
$table->string('name');
$table->string('email');
```

Text/string values [redacted] string column [redacted].

15. $table->integer()

```
$table->integer('age');
```

Whole-number values-[redacted] integer column [redacted].

16. $table->timestamps()

```
$table->timestamps();
```

[redacted] [redacted] **created_at** [redacted] **updated_at** columns [redacted].

17. down() for students

```
public function down(): void
{
    Schema::dropIfExists('students');
}
```

Migration rollback [redacted] students table-[redacted] remove [redacted] reverse operation.

18. Run Migration

```
php artisan migrate
```

Pending migrations-■ execute ■■■■■■ database schema-■■ update ■■■■■■■■■.

```
Migration file
      ↓
php artisan migrate
      ↓
up() executes
      ↓
students table created
```

19. make:migration vs migrate

```
php artisan make:migration create_students_table
                  ↓
             Creates migration FILE

php artisan migrate
                  ↓
             Executes migration
                  ↓
             Creates/changes TABLE
```

Remember: migration file ■■■■■■■■■ table create ■■■■■. `php artisan migrate` run ■■■■■ ■■■■■■■■.

20. Migration Status

```
php artisan migrate:status
```

Migration run ■■■■■■■■■■■■■■■ ■■■■■■■ pending-■ ■■■■■■■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■.

21. Rollback

```
php artisan migrate:rollback
```

Last migration batch-■ rollback ■■■■■■■■■. ■■■■■ migration-■■ `down()` method reverse operation-■ define ■■■■■■■■■.

22. Reset

```
php artisan migrate:reset
```

All migrations-■ rollback ■■■■■■■■■.

23. Refresh

```
php artisan migrate:refresh
```

Migrations-■ rollback ■■■■■■■ ■■■■■■■■■ run ■■■■■■■■■.

24. Fresh

```
php artisan migrate:fresh
```

Database-■■ ■■■■■ tables-■ drop ■■■■■■■ ■■■■■■■■ migrations-■ fresh-■ run ■■■■■■■■■■. Existing data delete ■■■■■■ ■■■■■■■■■ development/testing-■■ ■■■■■■■ ■■■■■■■■■■■ ■■■■■■■■■.

25. Migration Tracking

Laravel migrations run ■■■■■■■ track ■■■■■■ database-■■ **migrations** table ■■■■■■■■■■■■■■■■■■. ■■■■■■■ ■■■■■ migrations apply ■■■■■■■■■■■■■■■■■■ ■■■■■■■ Laravel ■■■■■ ■■■■■■■■■.

26. Complete Day 8 Practical

```
# MySQL running ■■■■■■■ ■■■■■■■■■

# Create database in MySQL
CREATE DATABASE firstproject;

# Configure .env
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=firstproject
DB_USERNAME=root
DB_PASSWORD=

# Create migration
php artisan make:migration create_students_table

# Edit migration
$table->id();
$table->string('name');
$table->string('email');
$table->integer('age');
$table->timestamps();

# Run migration
php artisan migrate

# Check status
php artisan migrate:status
```

27. Expected students Table

```
students
--------------------------------
id
name
email
age
created_at
updated_at
```

28. Full Mental Model

```
MySQL
  ↓
Create Database
  ↓
.env
  ↓
Laravel DB Configuration
  ↓
make:migration
  ↓
Migration File
  ↓
```

```
Define Table + Columns
  ↓
php artisan migrate
  ↓
MySQL Table
```

29. Quick Revision

```
DB_DATABASE      → Which database?
Migration         → What table structure?
make:migration    → Create migration file
up()              → Apply schema change
down()            → Undo schema change
Schema::create()  → Create table
$table->id()      → ID column
string()          → String column
integer()         → Integer column
timestamps()      → created_at + updated_at
migrate           → Execute pending migrations
status            → Check migration status
rollback          → Roll back last batch
reset             → Roll back all
refresh           → Roll back + migrate
fresh             → Drop tables + migrate again
```

30. Key Points to Remember

1. `.env` database connection details-■ configure ■■■■■■■■■.
2. `.env`-■■ database name ■■■■■■■■■ database create ■■■■■.
3. `make:migration` migration file ■■■■■■■ create ■■■■■■■■■.
4. `php artisan migrate` migration-■ execute ■■■■■■■■■.
5. `up()` apply change; `down()` reverse change.
6. Migration ■■■■■ database schema-■■ code-■ manage ■■■■■■■■■.
7. `students` migration-■■ `id`, `name`, `email`, `age`, timestamps define ■■■■■■■■■.